Lab 04 — Render Deployment (Summary)

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Result

A Node.js/Express web app was built locally, pushed to GitHub and deployed on Render. It is live, and every push to GitHub redeploys it automatically.

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Deliverable	Link
GitHub repository	github.com/cheihkaouimahdi-hub/Render
Live Render URL	render-262g.onrender.com

What was done

1. **Part 1 — Local app:** a Node.js project was created, Express was installed and a small server was written that replies "Hello from Render!". It ran locally on port 3000.
2. **Part 2 — GitHub:** the code was committed and pushed to GitHub. A .gitignore file kept node_modules and .env files out, so no secrets were published.
3. **Part 3 — Render:** a free Web Service was created from the GitHub repository with the Node runtime, "npm install" as the build command and "npm start" as the start command. The build took 28.9 seconds and the app went live on port 10000, which Render chose.
4. **Part 4 — Update:** the message was changed to "Hello from Render!/i am cheikhaoui ahmed mahdi" and pushed. Render detected the new commit, redeployed automatically in about one minute, and the live page showed the new message.

Answers to the questions

1. What does Render manage that a VM would leave to us? The server, the operating system and its security updates, the Node.js runtime, the build and deployment, restarting the app if it crashes, networking, the public URL with HTTPS, logs and monitoring, and scaling.

2. Roles of Node.js and Express. Node.js is the runtime that executes our JavaScript on the server. Express is a web framework on top of Node.js that creates the HTTP server, defines the routes and sends the responses.

3. Why use the PORT environment variable instead of only 3000? Render picks the port at runtime and gives it to the app through the PORT variable; here it was 10000. If the app only listened on 3000, Render could not route traffic to it. Port 3000 stays as a fallback for running locally.

4. Role of GitHub. GitHub stores the code that Render deploys. Each push notifies Render, which rebuilds and redeploys automatically. The commit history also allows rolling back to an earlier version.

5. Why this is PaaS. We supplied only the code and two commands; Render provided and ran everything underneath it. We never created, logged into or configured a server, which is what we would do with IaaS.

Evidence

Figure 1 — Local run at http://localhost:3000: *screenshot to add.*

Figure 2 — Render deploy log: build successful, service live.

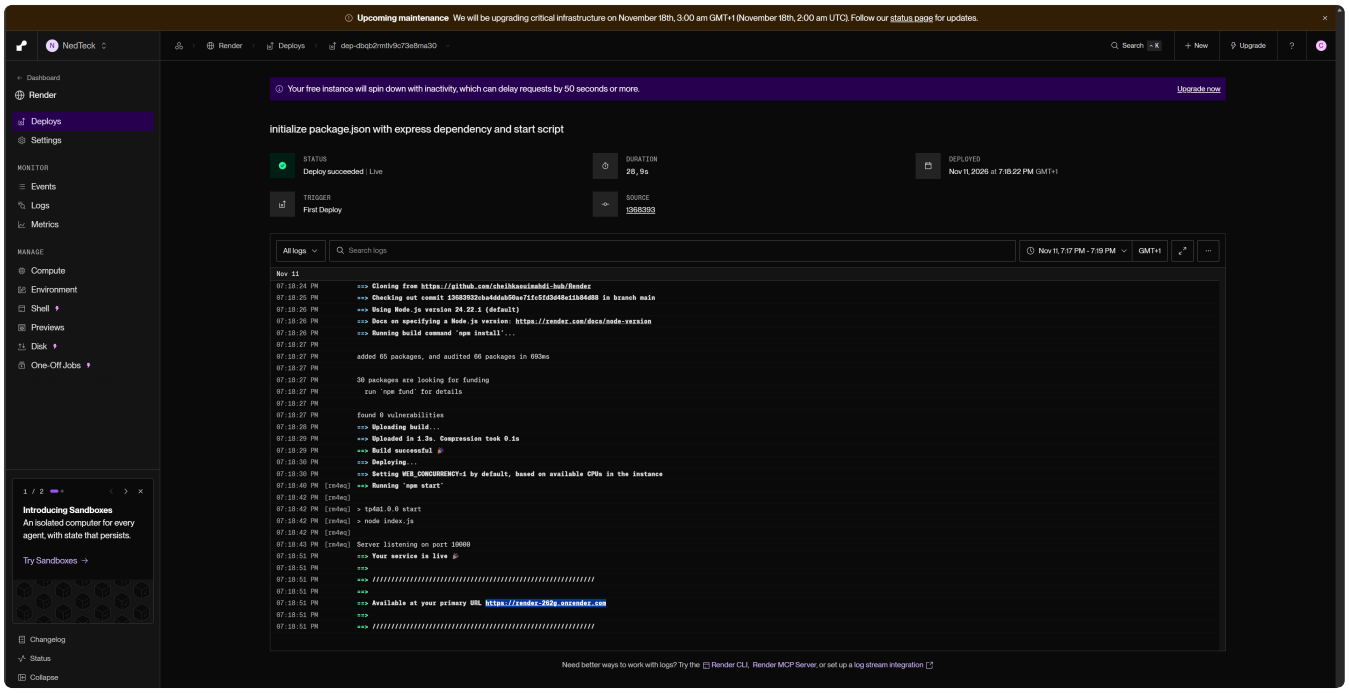


Figure 3 — Live app on Render showing "Hello from Render!".

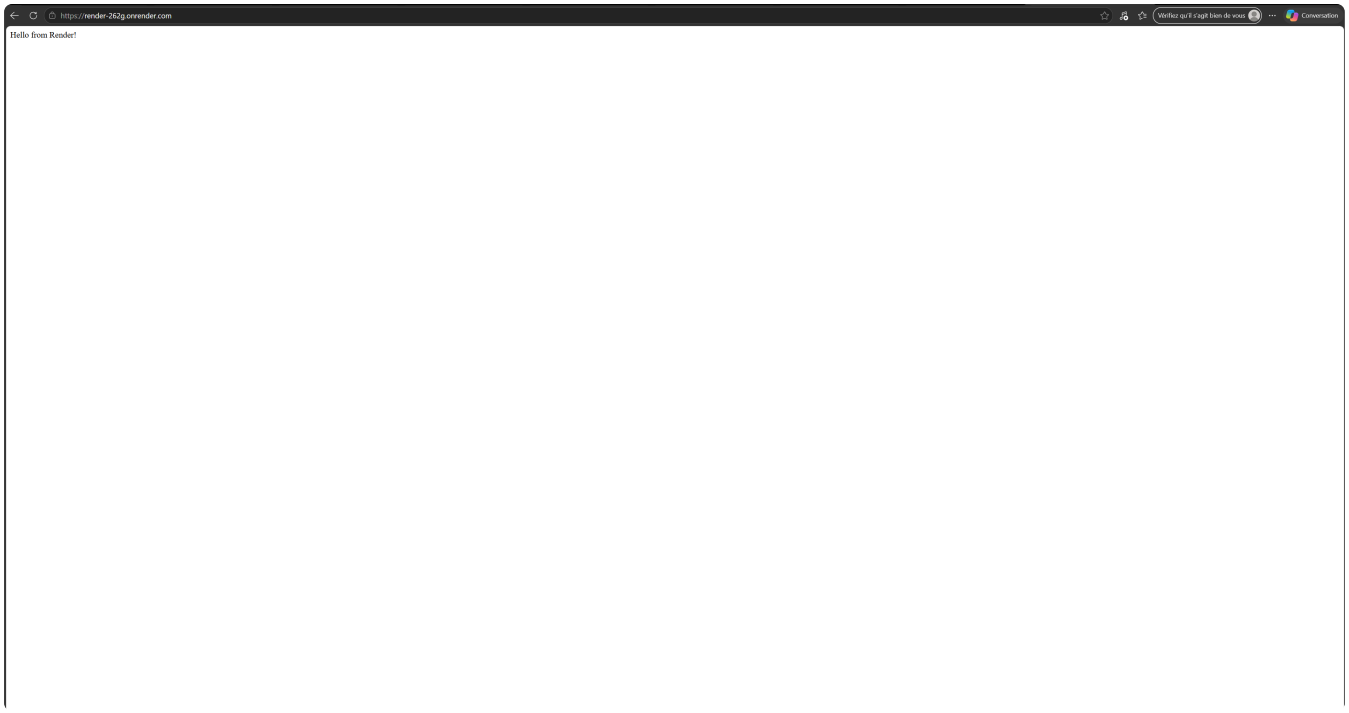


Figure 4 — Render Events: the push triggered an automatic redeploy.

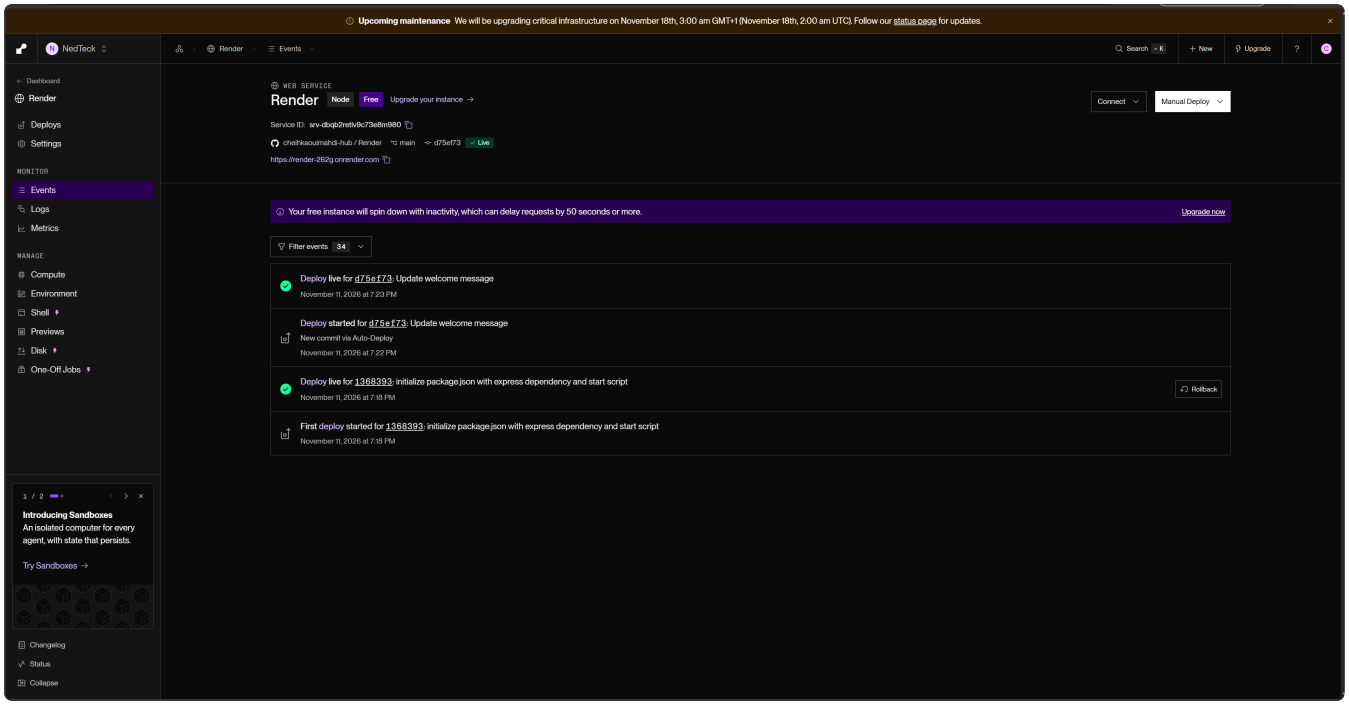
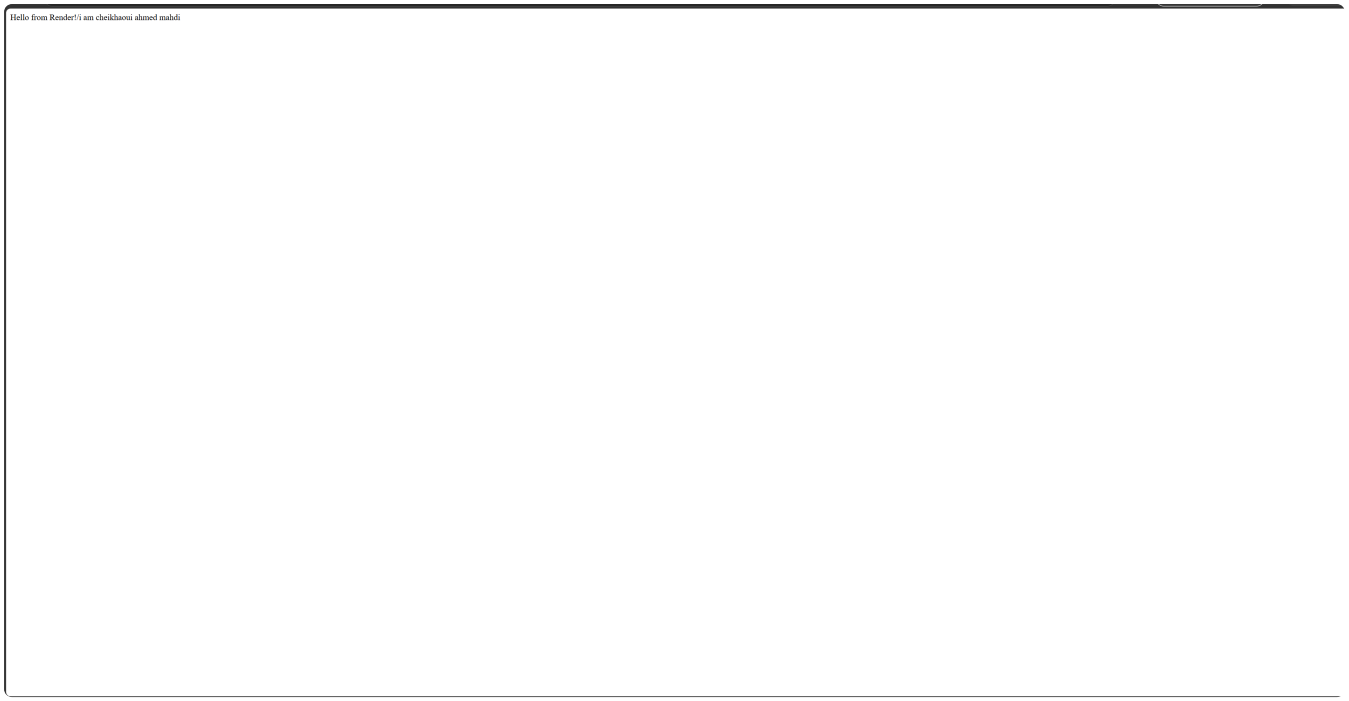


Figure 5 — Live app after the update showing the new message.


Hello from Render!/i am cheikhaoui ahmed mahdi